Supplementary Material: Stage-Wise Attribution of Geometric Error in Automatic Anatomical Surface Reconstruction From Computed Tomography

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***Abstract*—This supplement holds the analyses behind the revised paper that do not fit its six pages. Sections S1–S12 each answer one point of the review of the first version and name it; S13 and S14 report two checks made since, on the engine's repeatability and on the accuracy estimate the application derives from these results. Every table and figure is generated from the per-mesh results by `research/supplement_stats.py`; the paper's own tables come from `research/paper_stats.py`. Organ: spleen (MSD Task09, 41 scans) or the held-out pancreas (MSD Task07, 41 scans drawn at random); TS: TotalSegmentator in fast (3 mm) or full (1.5 mm) mode; canonical: no blur, smoothing or decimation; production: the application's settings.**

S1. Design and counts

Review: clarify design and counts. Table S1 counts the units, configurations, engine calls and meshes of every run. Configurations per mask are the 48-point grid (blur $\{0, 0.5, 1\}$ mm $\times$ Taubin $\{0, 10, 30, 60\}$ $\times$ decimation $\{0, 0.25, 0.5, 0.75\}$), production, and production with blur held at 0.5 mm (production-mm). Engine calls count one segmentation per scan and resolution level a level actually resampled; levels that leave a scan unchanged reuse its native mask. The meshes count every scored mesh; level effects use only the scans a level resampled (spleen: 41 at `iso1.5` and `iso3`, 14 at `slice5`; pancreas: 41, 41 and 35).

Which comparisons use which settings. Level effects, Table II of the paper and the held-out tests H1, H4 and H5 use the canonical configuration. Variance shares (Table III, H2, H3) use the 48 grid configurations. The grid comparison (H6) pairs native and `iso3` meshes at identical physical settings across all 48 grid configurations. Closure in Table II and the deployed-settings results use production; S7 compares production with production-mm.

S2. Tier 1 floor and smoothing by organ

Review: phantom interactions may not generalise. Table S2 is the canonical error floor on exact surfaces (Table II of the first version). Table S3 tests the interaction the phantoms showed, smoothing that helps compact shapes and harms the thin capsule, on both organs. On anatomy smoothing moves the native expert mesh away from its unsmoothed reference, as it must; on the 3 mm grid light smoothing slightly helps the pancreas expert mask (0.456 to 0.430 mm at 10 iterations) but not the spleen's, heavy smoothing harms both, and engine meshes change by at most 0.19 mm. On the phantoms the reversal between shapes holds without blur or decimation (0.271 to 0.160 mm for the sphere, 0.331 to 0.629 mm for the capsule at 2 mm). The interaction is therefore shape-specific: no organ showed the capsule's doubling of error, although the pancreas is the more sensitive of the two.

S3. Per-scan distributions and failure cases

Review: report per-scan distributions and failure cases, not only means. Fig. S1 shows every scan's canonical ASSD across levels and Table S4 its median, interquartile range and worst case. Table S5 lists the three worst scans per engine at native resolution, with the largest connected region the engine missed or added, as a share of the expert organ. The spleen failures are single missed regions (18–20% of the organ in scan 44); the pancreas failures are missed tumour: scan 415 holds 732 cm^3 of tumour around 135 cm^3 of pancreas, the engine segments the pancreas alone (Dice 0.16), and the merged reference therefore counts the tumour as missed organ. One spleen scan (26) is cut by the field of view; no pancreas scan is. Fig. S2 shows where the fast engine's error lies on a median spleen scan and on the worst.

S4. Sampling models and prediction intervals

Review: refine the empirical model; validate slice-only coarsening; report prediction intervals. Table S6 compares the paper's model M0 with three refinements on the spleen's 96 resampled expert masks, by leave-one-scan-out error, a 95% prediction interval (the 95th percentile of the absolute leave-one-scan-out residuals), the error when `slice5` is held out of the fit, and transfer to the held-out pancreas and to the box slice profile. A free intercept (M1) fits slightly better but cannot be trusted outside the fitted range: holding out `iso1.5` it misses by 0.28 mm. Linear axis terms (M2) fit worse, and quadrature axis weights (M3) converge to equal values (0.031 and 0.032), so the pooled Δ is the right form. M0's prediction interval of ± 0.111 mm covers 98% of pancreas points, and its slope is 0.177 [0.171, 0.184] on the spleen, 0.180 [0.176, 0.184] on the pancreas and 0.174 [0.167, 0.181] with the box profile. Slice-only coarsening remains the weakest prediction (0.109 mm when `slice5` is held out), because the spleen has 14 such scans and they span 1.5–4 mm native slices.

TABLE S1
DESIGN AND COUNTS OF EVERY RUN.

Tier	Data	Units	Reference	Segmentation	Levels	Configs	Engine calls	Meshes
Tier 1	phantoms	4 shapes × 5 placements	exact surface	–	4	97	–	7,760
Tier 2	Spleen	41 scans	expert mask	Expert mask, TS fast, TS full	4	50	137, 137	24,600
	Pancreas	41 scans	expert mask	Expert mask, TS fast, TS full	4	50	158, 158	24,600
Sensitivity	Spleen	41 scans	expert mask	Expert mask, TS fast	4	3	110	984

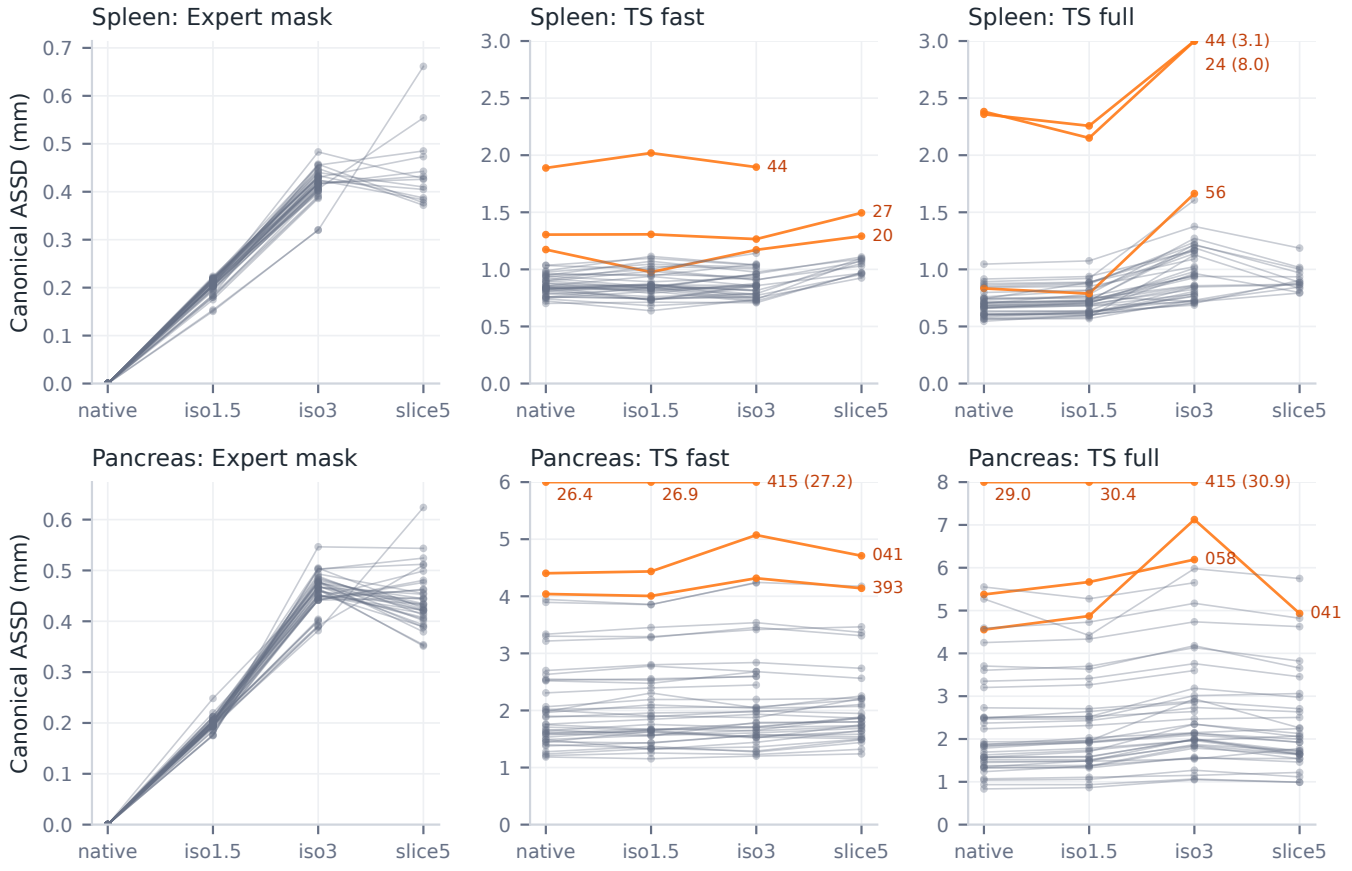


Fig. S1. Canonical ASSD of every scan across resolution levels, by organ and mask source. Orange: the three scans with the largest error; numbers in brackets are values above the axis.

TABLE S2
TIER 1 ERROR FLOOR: ASSD (HD95) IN MM OF THE CANONICAL RECONSTRUCTION AGAINST THE EXACT SURFACE, MEAN OF FIVE PLACEMENTS.

Phantom	0.5 mm	1 mm	2 mm	0.8 × 0.8 × 5 mm
Sphere	0.070 (0.17)	0.147 (0.34)	0.271 (0.75)	0.495 (1.62)
Capsule	0.069 (0.17)	0.144 (0.35)	0.331 (0.77)	0.634 (1.95)
Torus	0.068 (0.16)	0.138 (0.33)	0.278 (0.66)	0.499 (1.61)
Rounded box	0.072 (0.17)	0.145 (0.33)	0.297 (0.68)	0.525 (1.72)

S5. THE SLICE5 SUBSETS

Review: the 14-scan subset limits power; report per-scan effects and sensitivity. Table S7 gives each `slice5` effect with its leave-one-scan-out range, the number of scans whose error rose, and the minimum effect detectable with 80% power at $\alpha = 0.05$ (paired t approximation). Every spleen scan rose for every source, no single scan moves an effect by more than 0.02 mm, and the observed effects are about three times the detectable minimum. The pancreas, with 35 resampled scans, replicates the effect.

S6. TOPOLOGY WITHOUT THE FIELD-OF-VIEW-CUT SCAN

Review: one cut scan caps the topology curves. Table S8 repeats the closure rates with spleen scan 26 removed. Without decimation every mesh is then closed; the effect of decimation on closure is unchanged.

S7. BLUR IN VOXELS AND IN MILLIMETRES

Review: production blur is in voxels, research blur in millimetres. Table S9 compares production (0.6 voxels) with production-mm (0.5 mm, same smoothing and decimation) at every level. The two differ most on coarse grids, where 0.6

TABLE S3

MEAN ASSD (MM) BY TAUBIN ITERATIONS WITHOUT BLUR OR DECIMATION (LEWINER MARCHING CUBES FOR THE PHANTOMS).

Surface	Grid	0 it.	10 it.	30 it.	60 it.
Phantom: sphere	1 mm (exact)	0.147	0.126	0.068	0.064
	2 mm (exact)	0.271	0.223	0.160	0.174
Phantom: capsule	1 mm (exact)	0.144	0.127	0.107	0.141
	2 mm (exact)	0.331	0.322	0.629	1.599
Phantom: torus	1 mm (exact)	0.138	0.116	0.063	0.055
	2 mm (exact)	0.278	0.239	0.155	0.164
Phantom: rounded box	1 mm (exact)	0.145	0.125	0.068	0.086
	2 mm (exact)	0.297	0.260	0.206	0.320
Spleen: Expert mask	`native`	0.000	0.046	0.215	0.306
	`iso3`	0.418	0.422	0.475	0.537
Spleen: TS fast	`native`	0.901	0.911	0.961	0.978
	`iso3`	0.907	0.920	0.929	0.919
Spleen: TS full	`native`	0.790	0.813	0.869	0.890
	`iso3`	1.210	1.226	1.215	1.193
Pancreas: Expert mask	`native`	0.000	0.041	0.172	0.229
	`iso3`	0.456	0.430	0.437	0.553
Pancreas: TS fast	`native`	2.722	2.752	2.827	2.873
	`iso3`	2.842	2.865	2.914	2.961
Pancreas: TS full	`native`	3.021	3.057	3.142	3.174
	`iso3`	3.507	3.550	3.628	3.693

TABLE S4

CANONICAL ASSD ACROSS SCANS: MEDIAN [IQR] AND WORST SCAN (MM).

Organ	Source	Level	n	Median [IQR] (mm)	Worst (scan)
Spleen	Expert mask	`iso1.5`	41	0.207 [0.199, 0.216]	0.224 (63)
		`iso3`	41	0.418 [0.406, 0.432]	0.483 (13)
		`slice5`	14	0.427 [0.392, 0.466]	0.661 (53)
	TS fast	`native`	41	0.854 [0.810, 0.939]	1.888 (44)
		`iso1.5`	41	0.848 [0.808, 0.976]	2.019 (44)
		`iso3`	41	0.858 [0.781, 0.962]	1.896 (44)
		`slice5`	14	1.061 [0.968, 1.096]	1.495 (27)
	TS full	`native`	41	0.699 [0.627, 0.797]	2.382 (24)
		`iso1.5`	41	0.720 [0.634, 0.819]	2.257 (44)
		`iso3`	41	0.950 [0.790, 1.173]	7.966 (24)
		`slice5`	14	0.886 [0.850, 0.961]	1.186 (20)
Pancreas	Expert mask	`iso1.5`	41	0.199 [0.194, 0.204]	0.248 (075)
		`iso3`	41	0.461 [0.442, 0.478]	0.546 (339)
		`slice5`	35	0.433 [0.410, 0.462]	0.624 (120)
	TS fast	`native`	41	1.881 [1.532, 2.549]	26.418 (415)
		`iso1.5`	41	1.890 [1.574, 2.554]	26.919 (415)
		`iso3`	41	1.934 [1.572, 2.679]	27.187 (415)
		`slice5`	35	1.864 [1.643, 2.234]	4.709 (041)
	TS full	`native`	41	1.870 [1.476, 3.203]	28.976 (415)
		`iso1.5`	41	1.953 [1.492, 3.264]	30.383 (415)
		`iso3`	41	2.345 [1.869, 3.599]	30.904 (415)
		`slice5`	35	1.997 [1.643, 2.840]	5.749 (393)

voxels becomes 1.8 mm: on the expert masks by at most 0.065 mm, on the engines by at most 0.14 mm (TS full, pancreas, `iso3`). The paper's grid comparison (H6) already holds blur in millimetres.

S8. LOCAL AND TASK-SPECIFIC METRICS

Review: global metrics hide local defects; add tolerances. Table S10 adds NSD at 0.5, 1 and 2 mm, the larger directed 99th percentile (P99), the share of the delivered surface in local defects (faces more than 2 mm from the reference) with the largest connected patch (mm^2), the mean absolute volume error, and the share of meshes meeting two illustrative tolerances, HD95 $\leq$ 2 mm and |volume error| $\leq$ 5%. Resampled expert masks have defects only at `slice5`, on 0.2–0.7% of their surface; the engines have defects on 5–9% of the spleen surface and 27–29% of the pancreas surface, and on the pancreas fewer than 10% of engine meshes stay within 5% of the reference volume. The tolerances are examples, not validated clinical limits.

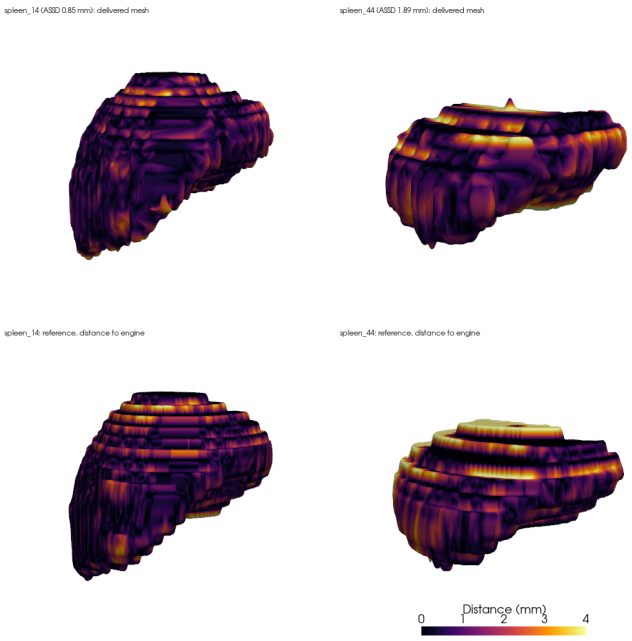


Fig. S2. Local error of TS fast on a median spleen scan (14) and the worst (44), coronal view. Top: the delivered production mesh coloured by its distance to the reference (added or displaced surface). Bottom: the reference coloured by its distance to the engine's mesh (missed anatomy); the bright block in scan 44 is the missed 18% of the organ.

S9. COARSENING-MODEL SENSITIVITY

Review: resampling does not reproduce real acquisitions. The spleen run was repeated at `iso3` and `slice5` with a rectangular slice profile one coarse voxel wide (a slab average, applied to the CT and the expert mask) and, for the engine, with white noise of SD 20 HU added to the Gaussian-coarsened CT (Table S11). The expert mask's sampling error changed by at most 0.003 mm, and the fast engine's error by at most 0.020 mm, about 2% of its value, with Dice unchanged. These bracket two simulation choices; they do not replace thin and thick reconstructions of the same acquisition, which this dataset does not contain.

S10. ENGINE MODES

Review: test full-resolution mode; is patient dominance fast-mode-specific? Table S12 pairs the two modes scan by scan. On the spleen, full mode is more accurate at native, `iso1.5` and `slice5` resolution but much worse on 3 mm input; on the pancreas it is worse at every level. Scans still dominate full mode's variance (spleen 59.3%, pancreas 99.4%; random intercept 59.6% and 99.5%), but on the spleen interactions between scan and resolution take a third of it, because the 3 mm penalty differs between scans. Resolution insensitivity is thus specific to the fast mode on the spleen, and scan dominance holds for both modes.

TABLE S5

THE THREE WORST SCANS PER ENGINE AT NATIVE RESOLUTION (CANONICAL MESH). MISSED AND EXTRA: LARGEST CONNECTED REGION THE ENGINE MISSED OR ADDED, AS % OF THE EXPERT ORGAN VOLUME.

Organ	Engine	Scan	Slice (mm)	Dice	ASSD (mm)	HD95 (mm)	Missed (%)	Extra (%)	Volume (%)
Spleen	TS fast	44	5	0.851	1.89	8.6	18.2	4.8	-9.7
		27	4	0.923	1.30	10.6	1.5	4.3	+1.0
		20	1.5	0.933	1.17	3.1	4.9	5.9	-0.0
Spleen	TS full	24	5	0.892	2.38	5.5	7.4	1.5	-9.1
		44	5	0.852	2.36	18.9	19.5	3.6	-13.7
		20	1.5	0.944	1.05	2.8	4.2	2.7	-1.1
Pancreas	TS fast	415	5	0.163	26.42	109.1	90.2	0.6	-89.7
		041	1.5	0.575	4.40	17.1	58.7	2.1	-56.5
		393	2.5	0.534	4.04	17.7	50.0	0.5	-61.7
Pancreas	TS full	415	5	0.137	28.98	116.7	91.7	0.5	-91.7
		365	5	0.647	5.55	41.8	35.0	0.9	-46.8
		058	5	0.546	5.38	33.2	60.7	2.0	-59.6

TABLE S6

SAMPLING MODELS FOR THE CANONICAL ASSD (MM) OF THE RESAMPLED EXPERT MASK. COEFFICIENTS IN MM PER MM (c IN MM^0); PI: 95% PREDICTION INTERVAL; PI COV.: SHARE OF PANCREAS POINTS INSIDE THE SPLEEN PI.

Model	Coefficients	R^2	LOSO MAE	95% PI	slice5 held out	Pancreas MAE (PI cov.)	Box profile MAE
M0: $b\Delta$	0.177	0.76	0.037	$\pm$0.111	0.109	0.030 (98%)	0.045
M1: $a + b\Delta$	0.066, 0.145	0.80	0.030	$\pm$0.093	0.099	0.030 (96%)	0.041
M2: $b_1\Delta_{\text{in}} + b_2\Delta_{\text{sl}}$	0.165, 0.159	0.54	0.056	$\pm$0.205	0.414	0.051 (98%)	0.067
M3: $\sqrt{c_1\Delta_{\text{in}}^2 + c_2\Delta_{\text{sl}}^2}$	0.031, 0.032	0.76	0.039	$\pm$0.120	0.254	0.030 (98%)	0.045

TABLE S7

PAIRED CHANGE IN CANONICAL ASSD FROM NATIVE TO `slice5` (MM). MDE: MINIMUM DETECTABLE EFFECT, 80% POWER.

Organ	Source	n	Change [95% CI] (mm)	Leave-one-out range	Scans up	SD	$\text{MDE}_{80\%}$
Spleen	Expert mask	14	+0.445 [+0.410, +0.489]	+0.429 to +0.451	14/14	0.079	0.064
	TS fast	14	+0.188 [+0.152, +0.228]	+0.175 to +0.195	14/14	0.078	0.063
	TS full	14	+0.212 [+0.173, +0.251]	+0.201 to +0.219	14/14	0.078	0.063
Pancreas	Expert mask	35	+0.442 [+0.425, +0.461]	+0.437 to +0.445	35/35	0.055	0.027
	TS fast	35	+0.143 [+0.116, +0.170]	+0.138 to +0.149	34/35	0.083	0.040
	TS full	35	+0.232 [+0.178, +0.285]	+0.221 to +0.240	32/35	0.162	0.079

S11. SCAN PROPERTIES BEHIND THE CASE EFFECT

Review: case difficulty is inferred, not measured. Table S13 correlates each scan's native canonical engine ASSD with measured properties: native spacing, expert organ volume, mean and SD of the organ's HU (eroded by one voxel), organ-to-surround contrast (organ mean HU minus that of a 3 mm soft-tissue shell around it), boundary-to-volume ratio and whether the organ touches the image edge; the last column is the leave-one-scan-out R^2 of a linear model on the continuous properties. Low contrast goes with higher error on both organs. On the spleen these properties explain little ($R^2 \approx 0.15$); on the pancreas, label volume, which here measures tumour burden, explains much more (R^2 0.59 and 0.76). The paper therefore calls the case effect between-scan variation and does not claim to have measured difficulty.

S12. SENSITIVITY ACROSS CONFIGURATIONS

Review: production settings are evaluated narrowly. Every engine conclusion was recomputed separately for each of the 49 configurations (48 grid points and production; Fig. S3, Table S14). Scans out-weigh resolution in every configuration for both organs and modes, and the `slice5` penalty is positive in every one. The fast engine's `iso3` change stays within ± 0.05 mm in 90% of configurations on the spleen and is positive in all of them on the pancreas; full mode's `iso3` penalty is at least 0.30 mm everywhere. The paper's conclusions therefore do not depend on the production settings.

S13. ENGINE REPEATABILITY

Since the revision: does the same scan give the same mask? Ten scans (five per organ, drawn with seed 2026) were segmented repeatedly in both modes: five times in one persistent GPU process, as the application runs the engine; three times in a new GPU process each, where nnU-Net's cuDNN autotuning may choose different kernels; and on the CPU, twice in fast mode and twice in full mode on one scan per organ. Each run is compared with the first run in the persistent process (Table S15). The persistent process returned

TABLE S8

SHARE OF CLOSED MESHES (%) OVER THE GRID BY DECIMATION: ALL SCANS / WITHOUT THE FIELD-OF-VIEW-CUT SCAN.

			Closed (%), all scans / without FOV-cut scans, by decimation			
Organ	Source	Level	0	0.25	0.5	0.75
Spleen	Expert mask	`native`	98 / 100	96 / 99	90 / 92	68 / 70
		`iso1.5`	98 / 100	95 / 97	92 / 94	75 / 77
		`iso3`	98 / 100	97 / 99	93 / 96	79 / 81
		`slice5`	100 / 100	100 / 100	93 / 93	71 / 71
Spleen	TS fast	`native`	98 / 100	97 / 100	93 / 95	57 / 59
		`iso1.5`	98 / 100	98 / 100	93 / 95	63 / 65
		`iso3`	98 / 100	95 / 97	94 / 96	83 / 85
		`slice5`	100 / 100	99 / 99	95 / 95	58 / 58
Spleen	TS full	`native`	98 / 100	96 / 99	89 / 91	58 / 59
		`iso1.5`	98 / 100	93 / 96	89 / 91	67 / 69
		`iso3`	98 / 100	94 / 96	92 / 94	82 / 84
		`slice5`	100 / 100	100 / 100	95 / 95	54 / 54
Pancreas	Expert mask	`native`	100 / 100	96 / 96	90 / 90	58 / 58
		`iso1.5`	100 / 100	98 / 98	93 / 93	71 / 71
		`iso3`	100 / 100	98 / 98	95 / 95	85 / 85
		`slice5`	100 / 100	100 / 100	94 / 94	66 / 66
Pancreas	TS fast	`native`	100 / 100	98 / 98	86 / 86	38 / 38
		`iso1.5`	99 / 99	95 / 95	86 / 86	51 / 51
		`iso3`	100 / 100	96 / 96	89 / 89	63 / 63
		`slice5`	100 / 100	94 / 94	88 / 88	49 / 49
Pancreas	TS full	`native`	100 / 100	94 / 94	75 / 75	38 / 38
		`iso1.5`	100 / 100	94 / 94	83 / 83	54 / 54
		`iso3`	100 / 100	96 / 96	90 / 90	64 / 64
		`slice5`	100 / 100	97 / 97	84 / 84	41 / 41

TABLE S9

PRODUCTION ASSD (MM) WITH BLUR IN VOXELS AND IN MILLIMETRES, PAIRED BY SCAN; CLOSED: % OF CLOSED MESHES, VOXEL / MM.

Organ	Source	Level	n	Voxel blur	mm blur	mm − voxel [95% CI]	Closed (%)
Spleen	Expert mask	`native`	41	0.234	0.226	-0.008 [-0.010, -0.006]	98 / 95
		`iso1.5`	41	0.374	0.358	-0.016 [-0.019, -0.013]	95 / 98
		`iso3`	41	0.517	0.486	-0.031 [-0.035, -0.028]	98 / 95
		`slice5`	14	0.425	0.417	-0.008 [-0.012, -0.004]	93 / 100
Spleen	TS fast	`native`	41	0.948	0.956	+0.008 [+0.004, +0.011]	66 / 51
		`iso1.5`	41	0.958	0.973	+0.016 [+0.011, +0.020]	90 / 90
		`iso3`	41	0.914	0.933	+0.018 [+0.011, +0.025]	95 / 98
		`slice5`	14	1.048	1.058	+0.009 [+0.004, +0.014]	50 / 64
Spleen	TS full	`native`	41	0.855	0.866	+0.010 [+0.004, +0.020]	98 / 93
		`iso1.5`	41	0.860	0.878	+0.017 [+0.007, +0.034]	93 / 88
		`iso3`	41	1.205	1.229	+0.024 [-0.026, +0.113]	95 / 85
		`slice5`	14	0.899	0.907	+0.008 [+0.004, +0.013]	86 / 86
Pancreas	Expert mask	`native`	41	0.189	0.182	-0.007 [-0.009, -0.006]	98 / 95
		`iso1.5`	41	0.293	0.277	-0.016 [-0.019, -0.013]	95 / 93
		`iso3`	41	0.535	0.470	-0.065 [-0.074, -0.056]	100 / 95
		`slice5`	35	0.423	0.399	-0.024 [-0.027, -0.021]	94 / 94
Pancreas	TS fast	`native`	41	2.836	2.837	+0.001 [-0.008, +0.011]	37 / 66
		`iso1.5`	41	2.911	2.887	-0.024 [-0.039, -0.010]	85 / 78
		`iso3`	41	2.977	2.931	-0.046 [-0.069, -0.022]	93 / 90
		`slice5`	35	2.268	2.257	-0.011 [-0.023, +0.002]	54 / 57
Pancreas	TS full	`native`	41	3.163	3.147	-0.016 [-0.033, +0.002]	76 / 71
		`iso1.5`	41	3.222	3.196	-0.026 [-0.039, -0.015]	93 / 80
		`iso3`	41	3.796	3.659	-0.137 [-0.194, -0.090]	90 / 80
		`slice5`	35	2.529	2.490	-0.039 [-0.065, -0.018]	77 / 74

identical masks in every repeat. A new process or the CPU changed at most 96 organ voxels, which moved the mesh by at most 0.025 mm and its ASSD against the expert surface by at most 0.014 mm, an order of magnitude below the between-scan differences in the paper. Meshing the same mask twice gave identical meshes for all 590 meshes of S14.

TABLE S10

LOCAL AND TASK-SPECIFIC METRICS (MEANS OVER SCANS; P99 IN MM; HD95 ≤ 2 AND $|\text{VOL}| \leq 5\%$: % OF MESHES MEETING THE TOLERANCE). CFG: CANONICAL OR PRODUCTION.

Organ	Source	Level	Cfg	$\text{NSD}_{0.5}$	NSD_1	NSD_2	P99	Defect (%)	Max patch	|Vol| (%)	HD95$\leq$2	|Vol| $\leq$5%
Spleen	Expert mask	native	prod.	0.88	1.00	1.00	0.87	0.0	0	0.3	100	100
		iso1.5	cano.	0.90	1.00	1.00	0.78	0.0	0	0.1	100	100
		iso1.5	prod.	0.73	0.94	1.00	1.38	0.0	0	0.7	100	100
		iso3	cano.	0.64	0.92	1.00	1.54	0.0	0	0.6	100	100
		iso3	prod.	0.60	0.86	0.99	1.96	0.0	0	2.0	93	98
		slice5	cano.	0.70	0.85	0.99	1.90	0.7	756	0.8	100	100
		slice5	prod.	0.72	0.88	0.99	1.92	0.5	452	1.0	93	100
Spleen	TS fast	native	cano.	0.41	0.66	0.89	4.98	8.9	2188	2.9	0	80
		native	prod.	0.40	0.64	0.89	4.87	5.0	1740	2.7	0	85
		iso1.5	cano.	0.42	0.65	0.90	4.78	6.5	2119	2.7	0	88
		iso1.5	prod.	0.38	0.64	0.90	4.76	3.3	1359	2.6	2	88
		iso3	cano.	0.41	0.65	0.90	4.79	4.3	1649	2.7	2	83
		iso3	prod.	0.38	0.66	0.91	4.54	1.8	1099	2.5	15	88
		slice5	cano.	0.31	0.58	0.88	5.32	8.8	2122	2.7	0	86
		slice5	prod.	0.32	0.59	0.88	5.15	6.2	1687	2.7	0	86
Spleen	TS full	native	cano.	0.47	0.74	0.93	6.03	5.0	1207	2.9	17	83
		native	prod.	0.45	0.70	0.92	6.05	3.1	950	2.8	12	83
		iso1.5	cano.	0.49	0.72	0.92	5.92	4.0	1056	2.9	15	85
		iso1.5	prod.	0.43	0.70	0.92	4.76	2.1	801	2.9	15	80
		iso3	cano.	0.38	0.61	0.87	8.03	5.8	2613	5.9	7	59
		iso3	prod.	0.33	0.59	0.86	6.75	3.1	1346	7.4	7	39
		slice5	cano.	0.36	0.64	0.91	3.69	5.0	804	3.9	0	79
		slice5	prod.	0.37	0.64	0.91	3.60	4.0	894	3.9	0	71
Pancreas	Expert mask	native	prod.	0.94	1.00	1.00	0.73	0.0	0	0.7	100	100
		iso1.5	cano.	0.92	1.00	1.00	0.73	0.0	0	0.4	100	100
		iso1.5	prod.	0.82	0.98	1.00	1.08	0.0	0	1.7	100	100
		iso3	cano.	0.61	0.92	1.00	1.43	0.0	0	2.2	100	98
		iso3	prod.	0.57	0.86	0.99	2.23	0.0	0	6.2	98	44
		slice5	cano.	0.67	0.81	1.00	1.39	0.2	322	0.8	97	100
		slice5	prod.	0.72	0.86	1.00	1.61	0.2	203	1.9	97	97
Pancreas	TS fast	native	cano.	0.23	0.41	0.64	15.64	29.2	12147	25.6	0	10
		native	prod.	0.21	0.38	0.63	15.92	24.0	8605	25.8	0	10
		iso1.5	cano.	0.23	0.39	0.63	15.60	27.0	10998	26.6	0	10
		iso1.5	prod.	0.20	0.36	0.62	16.01	21.7	8328	26.8	0	10
		iso3	cano.	0.20	0.38	0.64	15.90	22.6	9495	26.3	0	7
		iso3	prod.	0.19	0.36	0.61	16.31	18.0	8619	27.4	0	2
		slice5	cano.	0.18	0.35	0.65	12.01	26.5	9462	25.3	0	9
		slice5	prod.	0.18	0.34	0.62	12.28	24.7	8482	25.5	0	9
Pancreas	TS full	native	cano.	0.24	0.42	0.64	18.34	26.9	12191	30.8	0	7
		native	prod.	0.22	0.39	0.62	18.74	23.5	9898	31.1	0	7
		iso1.5	cano.	0.24	0.40	0.63	18.17	25.4	10482	30.8	0	7
		iso1.5	prod.	0.21	0.37	0.62	18.60	20.8	9111	31.5	0	10
		iso3	cano.	0.18	0.34	0.58	20.17	24.7	8152	34.4	0	12
		iso3	prod.	0.16	0.31	0.54	21.45	21.1	6704	37.2	0	12
		slice5	cano.	0.18	0.35	0.65	14.83	23.9	10864	29.7	0	9
		slice5	prod.	0.18	0.34	0.62	15.31	23.2	9235	30.2	0	9

TABLE S11

CANONICAL AND PRODUCTION ASSD (MM) UNDER THE GAUSSIAN COARSENING OF THE PAPER AND UNDER EACH VARIANT, PAIRED BY SCAN; DICE G / V: GAUSSIAN / VARIANT.

Source	Level	Variant	Config	n	Gaussian	Variant	Difference [95% CI]	Dice G / V
Expert mask	iso3	box	canonical	41	0.418	0.417	-0.001 [-0.002, -0.000]	0.968 / 0.968
		box	production	41	0.517	0.518	+0.001 [-0.001, +0.003]	0.968 / 0.968
	slice5	box	canonical	14	0.445	0.446	+0.001 [-0.001, +0.003]	0.971 / 0.971
		box	production	14	0.425	0.427	+0.003 [+0.001, +0.005]	0.971 / 0.971
TS fast	iso3	box	canonical	41	0.907	0.914	+0.006 [+0.003, +0.010]	0.940 / 0.940
		box	production	41	0.914	0.919	+0.005 [+0.000, +0.010]	0.940 / 0.940
		noise	canonical	41	0.907	0.926	+0.019 [+0.007, +0.031]	0.940 / 0.939
		noise	production	41	0.914	0.934	+0.019 [+0.008, +0.033]	0.940 / 0.939
	slice5	box	canonical	14	1.079	1.095	+0.016 [+0.004, +0.027]	0.926 / 0.925
		box	production	14	1.048	1.069	+0.020 [+0.006, +0.035]	0.926 / 0.925
		noise	canonical	14	1.079	1.081	+0.002 [-0.010, +0.014]	0.926 / 0.926
		noise	production	14	1.048	1.048	-0.000 [-0.015, +0.015]	0.926 / 0.926

TABLE S12

TotalSegmentator full against fast mode, paired by scan (ASSD in mm; full better: scans with lower full-mode ASSD).

Organ	Level	Config	n	Dice fast / full	ASSD fast	ASSD full	Full − fast [95% CI]	Full better
Spleen	`native`	canonical	41	0.940 / 0.951	0.901	0.790	-0.111 [-0.182, -0.009]	39/41
		production	41	0.940 / 0.951	0.948	0.855	-0.093 [-0.150, -0.017]	37/41
	`iso1.5`	canonical	41	0.940 / 0.951	0.905	0.805	-0.101 [-0.164, -0.008]	37/41
		production	41	0.940 / 0.951	0.958	0.860	-0.097 [-0.143, -0.047]	31/41
	`iso3`	canonical	41	0.940 / 0.932	0.907	1.210	+0.302 [+0.083, +0.689]	12/41
		production	41	0.940 / 0.932	0.914	1.205	+0.291 [+0.135, +0.545]	3/41
	`slice5`	canonical	14	0.926 / 0.937	1.079	0.913	-0.165 [-0.225, -0.121]	14/14
		production	14	0.926 / 0.937	1.048	0.899	-0.150 [-0.207, -0.105]	14/14
Pancreas	`native`	canonical	41	0.753 / 0.738	2.722	3.021	+0.298 [+0.087, +0.539]	18/41
		production	41	0.753 / 0.738	2.836	3.163	+0.327 [+0.118, +0.568]	17/41
	`iso1.5`	canonical	41	0.751 / 0.737	2.768	3.092	+0.324 [+0.102, +0.598]	15/41
		production	41	0.751 / 0.737	2.911	3.222	+0.311 [+0.084, +0.580]	14/41
	`iso3`	canonical	41	0.753 / 0.698	2.842	3.507	+0.665 [+0.417, +0.943]	5/41
		production	41	0.753 / 0.698	2.977	3.796	+0.819 [+0.549, +1.134]	5/41
	`slice5`	canonical	35	0.747 / 0.731	2.181	2.396	+0.215 [+0.051, +0.394]	13/35
		production	35	0.747 / 0.731	2.268	2.529	+0.261 [+0.073, +0.457]	14/35

TABLE S13

Spearman correlation of native canonical engine ASSD with scan properties (*: $p < 0.05$, uncorrected), and leave-one-scan-out R^2 of a linear model.

Organ	Engine	Slice spacing	In-plane spacing	Organ volume	Organ HU	Organ HU SD	Organ-to-surround contrast	Boundary-to-volume ratio	Cut by field of view	LOSO R^2
Spleen	TS fast	+0.16	-0.02	-0.03	+0.21	-0.05	-0.59*	+0.05	-0.09	0.14
	TS full	+0.20	-0.06	-0.14	-0.03	-0.13	-0.59*	+0.11	-0.01	0.15
Pancreas	TS fast	+0.20	+0.11	+0.49*	-0.36*	-0.01	-0.49*	-0.14	–	0.59
	TS full	+0.28	+0.20	+0.65*	-0.40*	+0.13	-0.44*	-0.17	–	0.76

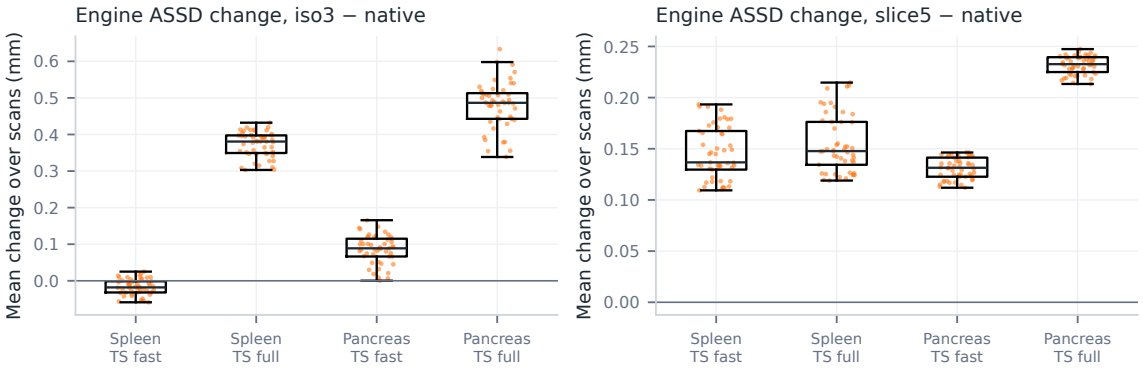


Fig. S3. Mean change in engine ASSD from native to `iso3` (left) and `slice5` (right), one point per reconstruction configuration.

TABLE S14

Engine conclusions across the 49 configurations: range of the mean ASSD change (mm), share of configurations with |change| ≤ 0.05 mm (`iso3`) or a positive change (`slice5`), and share where scans out-weigh resolution.

Organ	Engine	Configs	iso3 change range	|iso3| $\leq$0.05	slice5 change range	slice5 > 0	Case $>$ resolution
Spleen	TS fast	49	-0.059 to +0.025	90%	+0.109 to +0.193	100%	100%
	TS full	49	+0.303 to +0.432	0%	+0.119 to +0.215	100%	100%
Pancreas	TS fast	49	+0.000 to +0.166	20%	+0.112 to +0.146	100%	100%
	TS full	49	+0.338 to +0.633	0%	+0.213 to +0.247	100%	100%

S14. An accuracy estimate per mesh

Since the revision: can the results become an estimate shown with each mesh, and is it honest? The application reports three figures with every mesh: the resolution limit, $0.177\,\Delta$ with Δ the RMS voxel spacing and a range of 0.13–0.21 Δ from the Tier 1 floor; the shift its smoothing and decimation caused, measured against the mesh's own canonical surface; and, for the spleen and pancreas only, the engine's canonical

TABLE S15

REPEATED SEGMENTATION AGAINST THE FIRST RUN IN ONE PERSISTENT GPU PROCESS: RUNS, ORGAN VOXELS CHANGED AND ORGAN DICE (WORST RUN), MESH SHIFT (ASSD TO THE FIRST RUN'S MESH, MAX, MM), RANGE OF ASSD AGAINST THE EXPERT SURFACE ACROSS REPEATS OF ONE SCAN (MAX, MM), AND MEDIAN SECONDS PER SCAN.

Mode	Path	Runs	Organ voxels	Dice	Mesh shift	ASSD range	Seconds
fast	one persistent process	40	0	1.0000	0.000	0.000	6
	new GPU process	30	96	0.9993	0.017	0.014	30
	CPU	20	96	0.9990	0.025	0.014	34
full	one persistent process	40	0	1.0000	0.000	0.000	18
	new GPU process	30	16	0.9996	0.015	0.011	76
	CPU	4	4	1.0000	0.002	0.006	175

ASSD from this study as a median, interquartile range (IQR) and 90% range. Each was checked against the true error of the application's own meshes on all 82 scans, both modes and all four levels (590 meshes). On the held-out pancreas, 98% of coarsened expert surfaces fell within ± 0.111 mm of $0.177\,\Delta$ (mean miss 0.029 mm); every delivered mesh was at least as far from the expert surface as its resolution limit; the measured smoothing shift matched the change it caused in signed distance within 0.022 mm (median); and the watertight flag agreed with VTK's edge classification for all 590 meshes.

The engine figure was checked leave one scan out (Table S16, Fig. S4). Built from the native reference everywhere, with the paper's `slice5` change added on 5 mm grids (v1), its IQR held 46% of delivered meshes overall but only 15% of full-mode spleen meshes at `iso3`, where that mode degrades (S10). Taking the reference from the tested level closest to the scan's spacing (v2) brought this to 50% overall and at least 44% in every cell, with 86% inside the 90% range; empirical quartiles and 5–95% ranges of 40 scans are expected to hold a new scan 48% and 86% of the time. The estimate is thus calibrated within each organ on the scans that define it; it does not transfer between organs (the spleen's 90% range held 5% of pancreas meshes), and other structures are reported as not validated.

TABLE S16

SHARE OF DELIVERED MESHES (%) WHOSE ASSD AGAINST THE EXPERT SURFACE FELL INSIDE THE ENGINE FIGURE'S IQR AND 90% RANGE, LEAVE ONE SCAN OUT. V1: NATIVE REFERENCE EVERYWHERE; V2: REFERENCE FROM THE CLOSEST TESTED LEVEL.

Organ	Mode	Level	v1 IQR	v2 IQR	v1 90%	v2 90%
spleen	fast	`native`	49	56	90	90
		`iso1.5`	44	44	78	78
		`iso3`	37	46	71	76
		`slice5`	43	50	57	86
	full	`native`	61	63	93	93
		`iso1.5`	54	54	93	93
		`iso3`	15	46	56	90
		`slice5`	36	50	79	93
pancreas	fast	`native`	46	46	88	88
		`iso1.5`	49	46	85	85
		`iso3`	49	49	83	83
		`slice5`	54	51	86	83
	full	`native`	49	49	85	85
		`iso1.5`	51	51	88	85
		`iso3`	49	49	80	83
		`slice5`	46	51	86	89

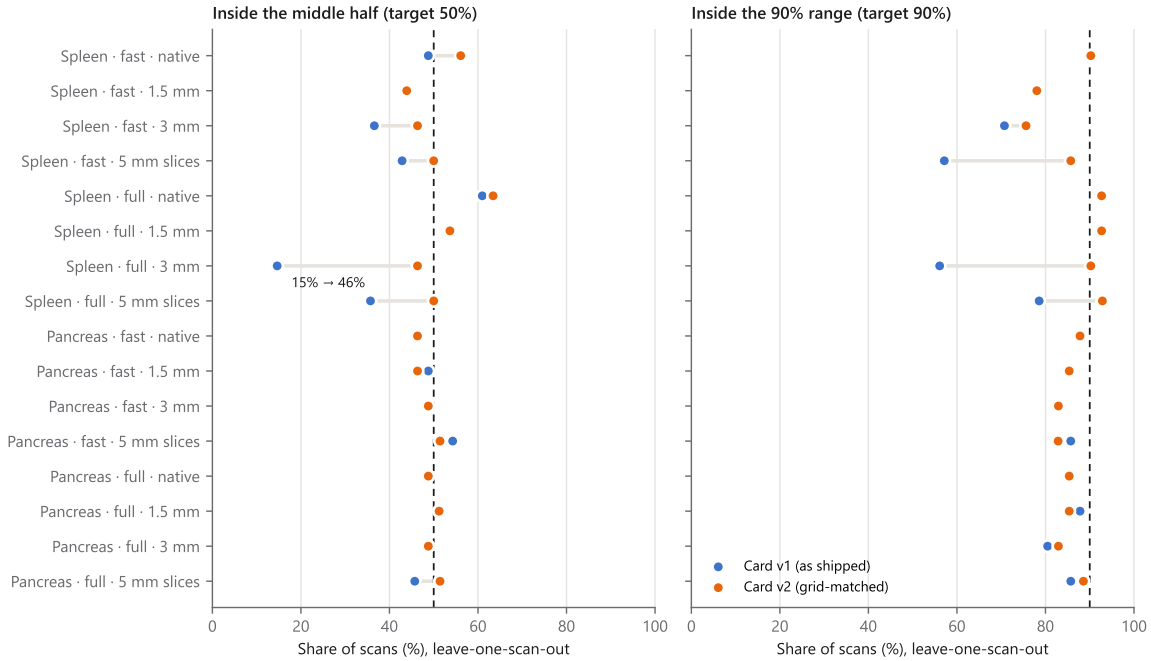


Fig. S4. Coverage of the engine figure before (v1) and after (v2) matching it to the scan's grid, by organ, mode and level; dashed lines mark the targets.